Delayed Distributed Coarsening in an Exactly Reversible Signed-Charge-Conserving Block Cellular Automaton

A computational companion to Zero Without Stasis*: delayed organisation and waiting-time kinetics under an exact signed balance*

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This is the computational companion to Zero Without Stasis*. It reports one finite formal example, a reversible artificial world whose signed total stays exactly zero while it organises, and grades every claim by the class of evidence behind it. A later causal inference from the same programme was withdrawn when its detector failed under intervention; the correction is recorded here rather than hidden. The abstract opens in plain language.*

Abstract

We found a simple artificial world in which random material reorganises into large, irregular material-rich and near-empty regions while the net signed charge stays exactly zero and every microscopic step can be undone. The result shows how substantial organisation can develop without breaking the global balance or discarding microscopic information; in this model, its timing depends strongly on initial density and its expression depends on the local reactions. We study a deterministic four-state block cellular automaton, whose cells are empty, positive, negative or bound-neutral, with a frozen 256-entry local permutation, selected in an exploratory search and then held fixed throughout the follow-up experiments. Each update conserves the signed charge $Q = N_+ - N_-$ (positive minus negative cells) but not the occupancy $M = N_+ + N_- + N_\mathrm{b}$ (all non-empty cells); the studied worlds start in the global $Q = 0$ sector. A 4096×4096 localised-patch trajectory developed irregular domains and, after 65,536 forward phases and the corresponding inverse phases, recovered all 16,777,216 cells exactly. Separate 1024×1024 homogeneous-random ensembles, with 12 seeds per density, exhibited density-sensitive domain formation with high local compositional disorder. No run at initial occupancy 0.16 crossed the fixed geometrical endpoint (a specified growth of connected empty regions, defined in Section 4) by 262,144 phases; at 0.18, 8/12 crossed by 524,288 phases. The Kaplan–Meier median first passage decreased from 495,616 phases at 0.18 to 65,536 at 0.26. Event-centred morphology favoured distributed change rather than a localised droplet front. In the pooled control protocol, an active sibling conserving both Q and M and a pure-rotation rule each gave 0/16 end-of-window strong-domain passes, versus 10/16 for the selected rule; several single reaction-pair removals also suppressed them. The contribution is this reproducible, quantitatively controlled finite-time regime, not reversibility or signed balance alone; the data are finite-time and do not establish an equilibrium phase transition.

Keywords: reversible cellular automata; Margolus block automata; signed charge conservation; coarsening; first passage; classifier correction

Plain-language significance Significance

Reversible rules do not prevent macroscopic structure, but dramatic pictures alone do not identify a mechanism. This study pairs an exact cell-for-cell reversal test with waiting-time ensembles, density sweeps, arrangement controls, a matched double-conservation sibling, and reaction-pair ablations. The central result is deliberately narrow: one fully specified reversible, signed-charge-conserving automaton produces delayed, large-scale, irregular organisation from random material, and that behaviour is strongly conditioned by density and by the reaction architecture tested. All trajectories remain exactly recoverable in principle and in the large direct audit.

1. Introduction

This paper is the computational companion to *Zero Without Stasis* [1], which proposes exactly this kind of experiment (its Section 14.1, *Signed Life*) and states in advance what a simulation can and cannot show for that framework. The present study stands on its own as a finite computational result; the parent paper's metaphysical claims are neither assumed nor tested here. The model is a finite, fixed-lattice realisation of that Signed Life programme; it does not implement the distinct splitting-and-merging refinement conjecture of the parent paper's Sections 6.4 and 14.2, which motivated the work.

Reversible cellular automata and lattice gases have long supplied constructive examples of local invertibility, conservation, and emergent statistical behaviour [2–6]. Conservative lattice gases recover transport and hydrodynamic phenomena [7–10]; reversible Ising and Q2R models display relaxation, long cycles, and phase-like macroscopic behaviour [11–15]; and reversible aggregation shows that invertible local dynamics can support cluster growth [16]. More recent one-dimensional reversible automata with vacancies and two particle species provide close alphabet, reaction, and conserved-charge precedents [17, 18], while reactive lattice-gas automata provide established examples of waves and pattern formation [19]. We therefore make no claim that reversibility plus organisation, multi-species reversible dynamics, or reaction-driven patterning is new in itself.

The narrower question is what can be established for a four-state signed system in which one additive quantity is exact while total occupancy can change. The distinction matters. A rule may conserve

$$Q = N_+ - N_-, \tag{1}$$

without conserving

$$M = N_+ + N_- + N_\mathrm{b}. \tag{2}$$

Writing "$Q = 0$" describes the global sector of the runs, not a second local conservation law and not an empty world. This paper asks whether a frozen rule (one selected in the exploratory search and then held fixed, never re-tuned, through all follow-up experiments) in that architecture reproducibly forms large irregular domains, how density and waiting time condition the outcome, and which nearby rule controls retain or suppress it.

Our terminology is descriptive. "Coarsening-like" refers to growth of long-wavelength structure and connected regions over the observed finite lattice and time window. "Strong-domain" refers to a fixed operational endpoint. Neither term asserts equilibrium phase separation, a critical point, a free-energy description, or a universality class [20, 21].

2. Model and exact audits

In plain terms, the world is a square grid of cells wrapped into a torus. Every cell is in one of four states: empty, positive, negative, or bound-neutral. Time advances in phases. In each phase the grid is cut into 2×2 blocks, with the cut shifted diagonally by one cell on alternate phases so that no block boundary is permanent, and every block is replaced by looking its four-cell pattern up in a fixed table of 256 entries. The table is a permutation of those 256 patterns, so each phase can be undone exactly by looking up the inverse table. Two counts matter throughout. The signed charge Q, positive cells minus negative cells, is unchanged by every table entry: it is a hard law of this world, not a quantity corrected afterwards. The occupancy M, the number of non-empty cells, is free: some entries create or remove a balanced positive–negative pair, so material can be made and unmade while Q stays put. That freedom is what the matched controls of Section 5.4 probe.

2.1 Four-state partitioned dynamics

Each square-lattice cell takes a value in $\{0, +, -, \mathrm{b}\}$: empty, positive excitation, negative excitation, or bound-neutral excitation. Alternating 2×2 Margolus partitions tile the periodic lattice. The four input symbols of each block are packed into one of 256 block states; a 256-entry permutation maps that block state to an output state. Applying the inverse permutation in reverse phase order recovers the predecessor exactly. The frozen winner's raw unsigned-byte rule table has SHA-256

`d1fe99b5735d8e9be032347de2727dc5c52a7848df200ca26f67f28c8c1bc41c`.

All 256 local cases were exhaustively audited. For every block input, the permutation is one-to-one and preserves the block contribution to Q. The winner does not preserve the block contribution to M; bound/unbound and occupied/empty reaction channels allow occupancy to change while signed balance remains exact. The construction starts from clockwise block rotation and exchanges 12 reversible pairs of outputs, so 24 of the 256 input blocks differ from pure rotation. Figure 2 shows the alternating partition and two entries from the archived table, including an actual reaction in which a balanced positive–negative pair is added while Q is unchanged.

2.2 Large exact reversal

The audited 4096×4096 trajectory began with a random central square spanning 45% of each lattice dimension, with the rest empty. Within the patch the target state fractions were 12% positive, 12% negative and 8% bound-neutral, with exact positive–negative balance. It was advanced for 65,536 partition phases and then evolved backwards with the exact inverse schedule. The recovered lattice matched its initial state at all 16,777,216 cells. Global Q remained zero throughout; M changed substantially. This is an implementation and coexistence audit, not a claim that reversibility itself is novel.

3. Discovery, selection, and confirmation

3.1 A small, explicitly selective search

The winner was search-selected, not sampled blindly. The original discovery run, a small-scale exploratory search (a "smoke" run in the software's terms), evaluated 14 unique rules

(16 replicate evaluations) in a reactive-rotation construction family. Its declared heuristic rewarded intermediate bands of activity, reaction density, state and block entropy, spatial and temporal mutual information, recurrence, perturbation distance, and patch repetition, with a dynamics gate and a hard charge-preservation gate. Thus subsequent behaviour cannot be described as wholly unselected.

The discovery score did *not* use the later 4096^2 reversal result, density first-passage curves, event morphology, arrangement controls, double-conservation sibling, reaction-pair ablations, prevalence screen, or the later classifier diagnostics. Those were frozen and run as follow-up measurements. The exact search provenance and metric weights are retained with the release.

3.2 Separation of evidence classes

We use four labels. *Discovery* denotes the tiny search and exploratory smoke run. *Confirmation* denotes frozen winner repeats and the large exact reversal. *Mechanism constraints* denote preregistered density, geometry, sibling, and ablation comparisons. *Diagnostics* denote analyses that reveal an assay limitation and cannot support their original causal question. Results below state their class rather than pooling them as if they were exchangeable.

The later prevalence screen found 19 strong-domain rules among 256 independently generated charge-preserving candidates (7.4219%; Wilson 95% interval 4.8026–11.3001%). This shows that the operational regime is not unique to one table within the sampled generator, but it neither erases the original selection nor estimates prevalence in all 256! block permutations.

4. Measurements

The v0.3/v0.3.1 strong-domain endpoint is a geometrical, trajectory-prefix classifier; it does *not* include a spectral-growth threshold. In words, the flag fires once the finite (non-wrapping) empty regions have grown, at some past sample, to at least four times their initial characteristic scale, still stand at least twice that scale at the current sample, the largest such region at that peak sample covers at least 0.1% of the lattice, and the global charge has never changed. Formally, let L_j be the measured characteristic scale of finite empty components at sampled phase t_j, let $r_j = L_j/\max(L_0, 1)$, and let j^* be the first index attaining the largest r_j in the prefix through t_k. Writing f_j for the lattice fraction in the largest finite empty component, the implemented flag is

$$C_k = \mathbf{1}\left[r_{j^*} \geq 4, \quad r_k \geq 2, \quad f_{j^*} \geq 0.001, \quad Q_j = Q_0 \text{ for all } j \leq k\right]. \tag{3}$$

Here $\mathbf{1}[\cdot]$ equals 1 when every listed condition holds and 0 otherwise. A prefix requires at least two snapshots. Spectral power, interface density, anisotropy and spatial mutual information are *separate measurements*, not additional gates in Eq. 3. This distinction is essential to the later shuffled-state false positives.

For the v0.3.1 first-passage study, T_{hit} is the first sampled phase with $C_k = 1$. The separate persistent endpoint, $T_{\text{persistent}}$, is the first sample starting three consecutive true flags. Samples are 2,048 phases apart. A transient first hit need not be the persistent hit. Runs that had not crossed by the end of their follow-up (right-censored runs) are retained as such rather than assigned artificial event times, and Kaplan–Meier medians are reported only when reached [22, 23]. Earlier campaigns used related but distinct frozen endpoints; the Supplement states their differences rather than treating all reported pass counts as interchangeable.

All quantitative claims refer to the stored sizes, seeds, phase horizons, and frozen thresholds. The endpoint is useful for comparing these runs but is not a natural law. Exact local-state

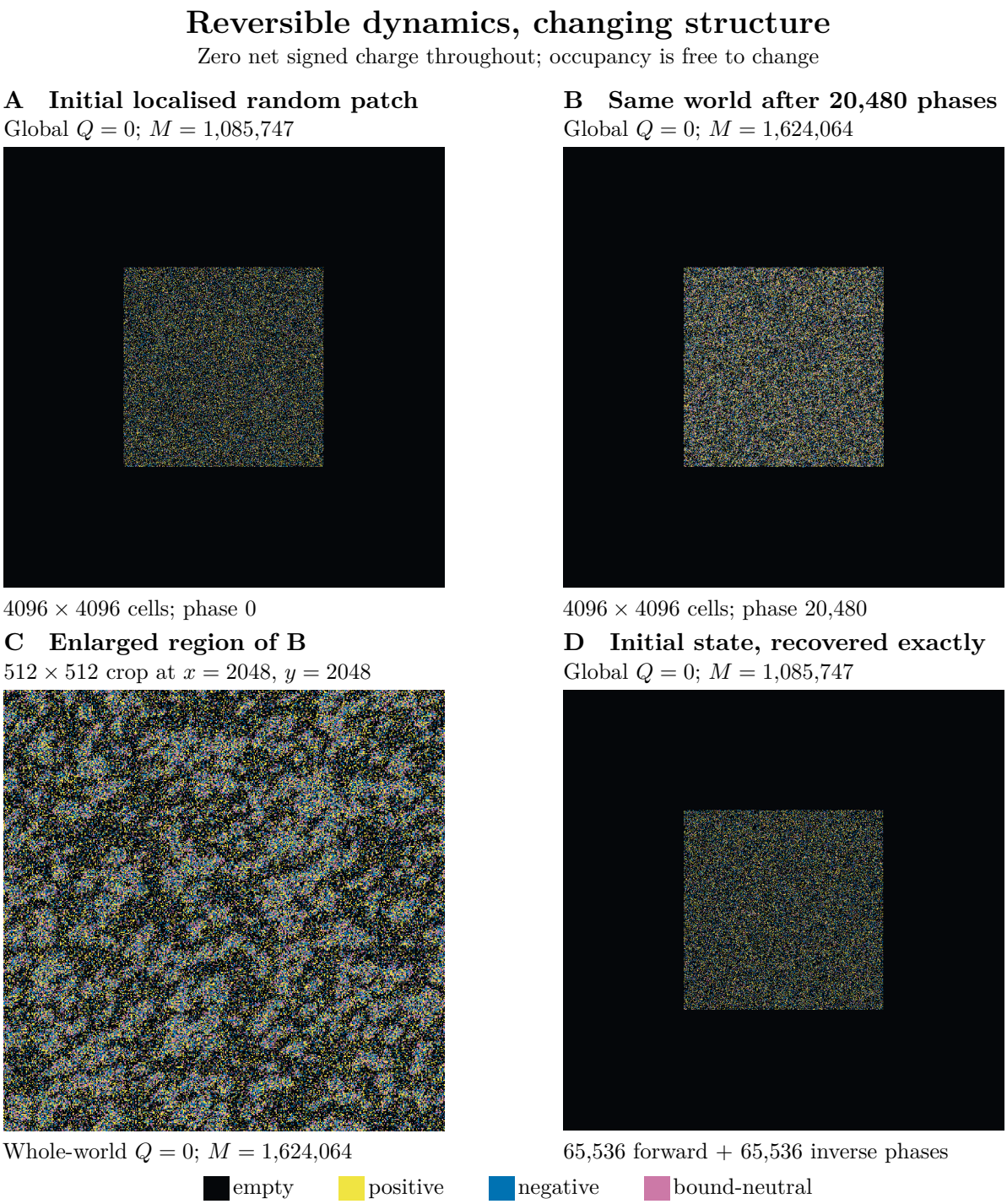


Figure 1: The world reorganises itself without ever leaving its exact zero balance, and every step of it can be undone. A and B are the same 4096 × 4096 world at phases 0 and 20,480: scattered random material to begin with, then broad excitation-rich regions separated by near-vacuum regions. C enlarges a 512 × 512 region of B, at the scale where the mixture inside a dense region can be seen to stay irregular rather than settling into a lattice or a stripe. D repeats the phase-0 reference state: a separate audit ran 65,536 phases forward and 65,536 back through the inverse table and recovered that state with zero mismatches across all 16,777,216 cells, which is why A and D are the same picture. Signed charge Q is zero in every panel, while occupancy M rises from 1,085,747 at phase 0 to 1,624,064 at phase 20,480: the balance is fixed, the amount of material is not. This is one localised-patch trajectory, seed 260824, and a different preparation from the homogeneous-random ensembles behind the density and control results.

images are rendered with nearest-neighbour sampling. Reported "reaction density" is the fraction of updated blocks that change reaction channel; it is not a chemical rate constant.

5. Results

5.1 Large-scale order with local compositional disorder

The localised-patch studies and the later whole-torus homogeneous-random studies both showed broad excitation-rich regions separated by near-vacuum regions, under their respective preparation and density conditions. The numerical composition result here comes from the four 4096×4096 localised-patch confirmation runs: the median occupancy of the dense interior (cells in the high-local-density band, excluding a margin around the dense/sparse interface) at the final stored snapshot of each run was 0.572989, with positive, negative and bound-neutral fractions 0.192542, 0.192535 and 0.187912. Normalised four-state compositional entropy was 0.946343. High single-site diversity is not itself proof of spatial randomness; the images and

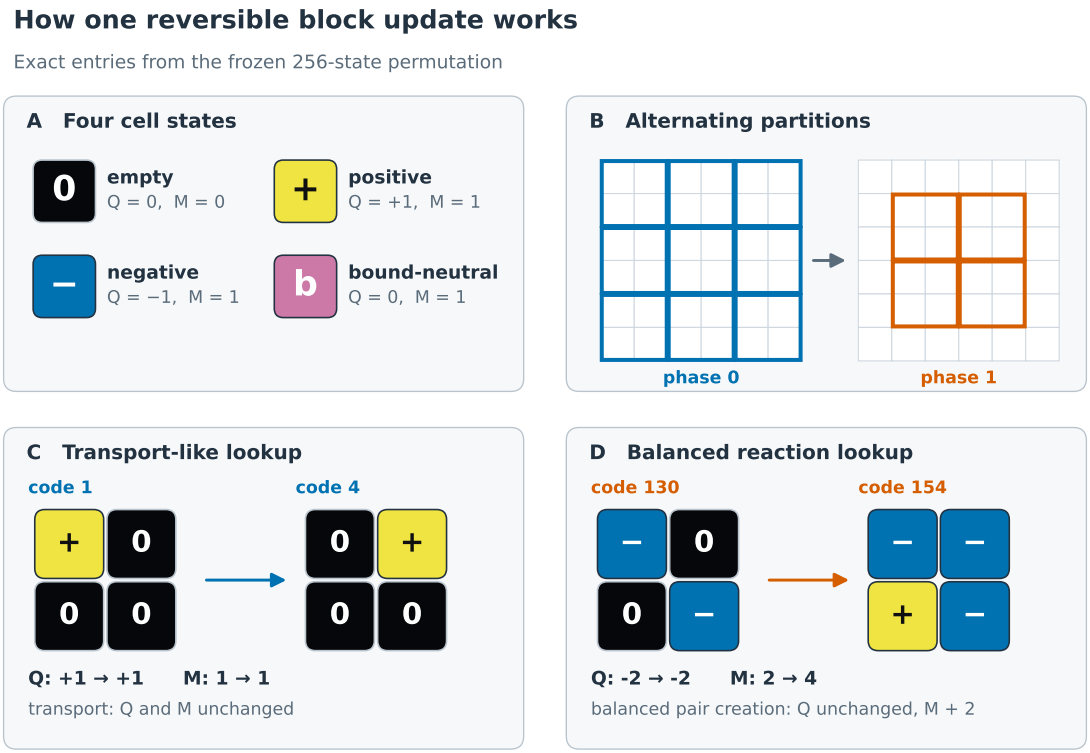


Figure 2: The whole dynamics is one lookup table, and this is what two of its entries do. A gives the four cell states with their contributions to the signed charge Q and the occupancy M; a bound-neutral cell counts towards M but not towards Q. B shows the 2×2 partition shifting diagonally between phases, so that no block boundary is permanent. C and D are exact entries from the frozen table, read top-left, top-right, bottom-left, bottom-right: entry $1 \mapsto 4$ simply moves an excitation, following the clockwise rotation the table was built from, while entry $130 \mapsto 154$ adds one positive and one negative cell together, taking M from 2 to 4 and leaving Q at -2. The table is a permutation of all 256 block patterns, built by exchanging twelve reversible output pairs in that rotation scaffold, so 24 of the 256 inputs differ from pure rotation and the inverse table sends 154 back to 130 exactly. Neither entry on its own accounts for the large-scale behaviour.

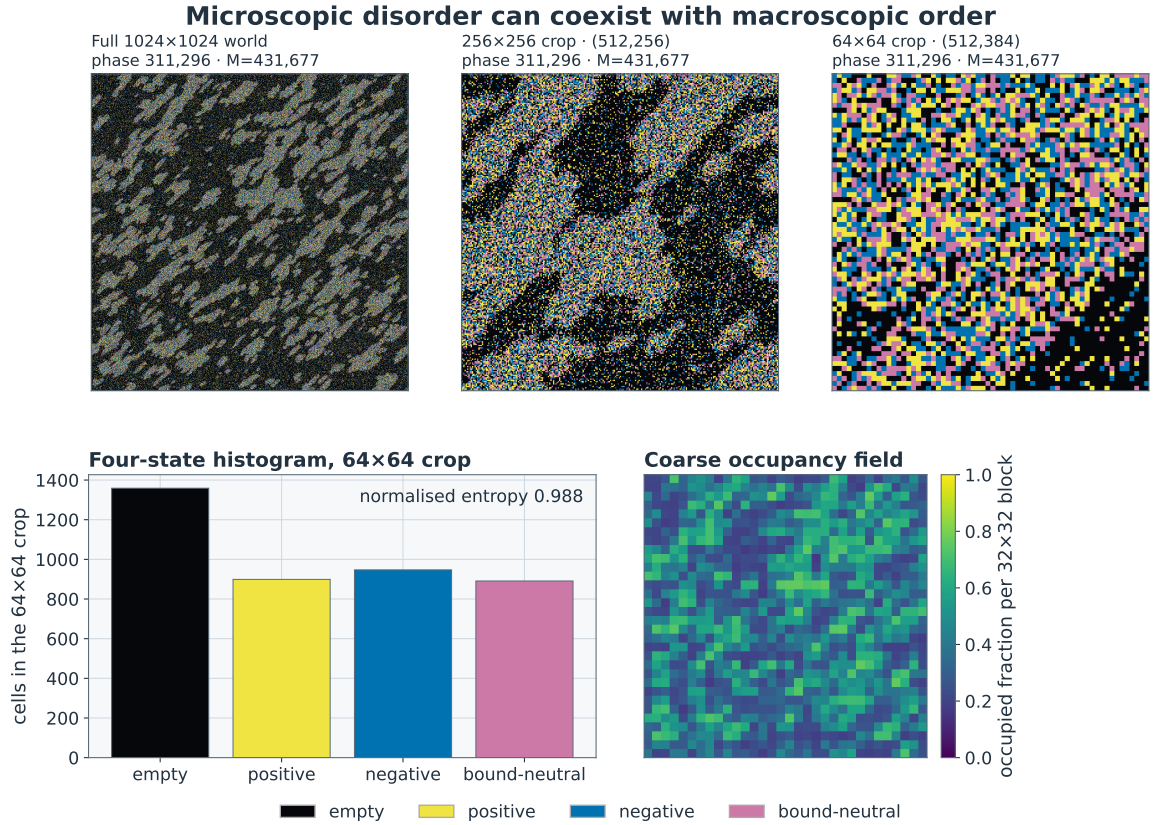


Figure 3: Order at the large scale is made from a mixture that stays thoroughly mixed up close. The top row zooms into one untouched homogeneous-start 1024^2 world at initial occupancy 0.20, seed 6203 and phase 311,296: the full torus, then two nested crops, down to a 64×64 crop in which individual cells are visible. The bottom row measures that same state two ways: the four-state histogram of the 64×64 crop, whose normalised entropy is 0.987741 out of a possible 1, and a coarse occupancy field averaged over 32×32 blocks, in which the large domains appear as smooth light and dark patches. So the domains are plain at the block scale with no checkerboard, stripe or crystal at the cell scale. This is a description of morphology, not a claim of equilibrium or of a phase transition, and high single-site entropy on its own does not exclude every kind of spatial order. It is a picture of one untouched source world, distinct from the four 4096^2 localised-patch composition averages quoted in Section 5.1, and it is not evidence about the withdrawn intervention study of Section 7.

separately recorded spectral diagnostics support the description of irregular, noncrystalline morphology.

Across those four localised-patch confirmatory winner runs, the transient domain-scale peak ranged from 72.66 to 136.11 times its initial value (median 98.4459), while the final recorded scale remained 15.77–16.33 times initial (median 16.1118). The effect is large but nonmonotone: the preregistered sustained late-time coarsening criterion (a separate, stricter test that also required continued growth of the empty-domain scale through the late half of the run; Supplement, Section 5.3) was not met at this scale. The large transient morphology is not a success of that stronger hypothesis.

5.2 Density controls waiting time

The first-passage cohort comprised homogeneous-random 1024×1024 worlds. At an initial occupancy of 0.16, none of 12 runs reached the endpoint within 262,144 phases. At 0.18, 8/12 reached it in the extended 524,288-phase campaign, with median first passage 495,616 phases. All 12 replicates reached it at each tested density from 0.20 through 0.26; median first passage

Table 1: Homogeneous 1024^2 first-passage cohort, with actual follow-up horizons.

Initial occupancy	Events / 12	KM median (phases)	Follow-up horizon (phases)
0.16	0	not reached	262,144
0.18	8	495,616	524,288
0.20	12	321,536	262,144 / 524,288[a]
0.22	12	206,848	262,144
0.24	12	108,544	262,144
0.25	12	83,968	262,144
0.26	12	65,536	262,144

Homogeneous 1024^2 worlds; Kaplan–Meier median is the first sample at which the estimated survival is at most 1/2, not the median of observed events alone. [a]At occupancy 0.20, two runs had crossed by 262,144 phases and kept that record; the other ten, still uncrossed at 262,144, were continued to 524,288 phases and all crossed. Four of the 0.18 trajectories remained right-censored at 524,288. Recorded horizons, not an imputed common duration, are shown.

fell from 321,536 phases at 0.20 to 65,536 at 0.26 (Table 1). Across 0.18 to 0.26 the medians lie close to a straight line on a logarithmic scale: each percentage point of starting density removed lengthens the median wait by a factor of about 1.3 (an unweighted least-squares fit of log median phase against initial occupancy, six points, $R^2 = 0.995$; descriptive, not a law). Within the v0.3/v0.3.1 endpoint, longer observation admitted events at densities that rarely crossed at shorter times. The earlier approximately 27.5% onset came from a related but different frozen classifier and is not a numerically interchangeable threshold estimate. The defensible result is strong finite-time density dependence, not an equilibrium critical density.

5.3 Distributed events and the prepared-geometry comparison

Each endpoint crossing was examined in a frozen window of the samples leading up to it and labelled by where the change happened: *localised* if at least half of the sites whose occupancy changed lay inside one fixed crop covering a sixteenth of the lattice (a single growing nucleus), *distributed* if that crop held no more than an eighth of the changed sites while the finite-empty scale grew in at least half of the pre-hit intervals, and *unresolved* otherwise (two further preregistered labels, merger and reconnection, matched no event). The pool is the 91 first hits observed in the v0.3.1 study, 68 from the primary 1024^2 cohort and 23 from the 512^2 area-scaling cohort. The labels classified 67 transitions as distributed, 24 as unresolved, and none as localised droplet-front events. Under a simple local Poisson picture, in which independent nucleation events occur at a fixed rate per cell, the first-event rate per cell would be the same on the matched 512^2 and 1024^2 lattices; instead the smaller lattice showed roughly four times the per-cell rate at every matched density, so that model was not supported. These finite-window observations constrain a localised-front interpretation without excluding every nucleation description or establishing an asymptotic rate law.

Prepared starts that place the same occupied inventory in a single compact disc (a droplet) or a single full-height band (a slab), at a set internal density with the rest of the torus empty, produced 0/48 endpoint crossings by 262,144 phases, versus 19/24 for matched homogeneous starts, while their imposed occupied structures persisted or grew. Crucially, Eq. 3 measures finite empty-component growth relative to each trajectory's own initial finite-empty-domain scale; it is not an occupied-seed survival test. Preparing a large occupied shape changes the starting geometry and its normalization baseline relative to random soup. The pass-count contrast is not a common absolute test of whether a large structure forms or survives, and it cannot by itself show that seeded organisation fails or that compact seeds inhibit the phe-

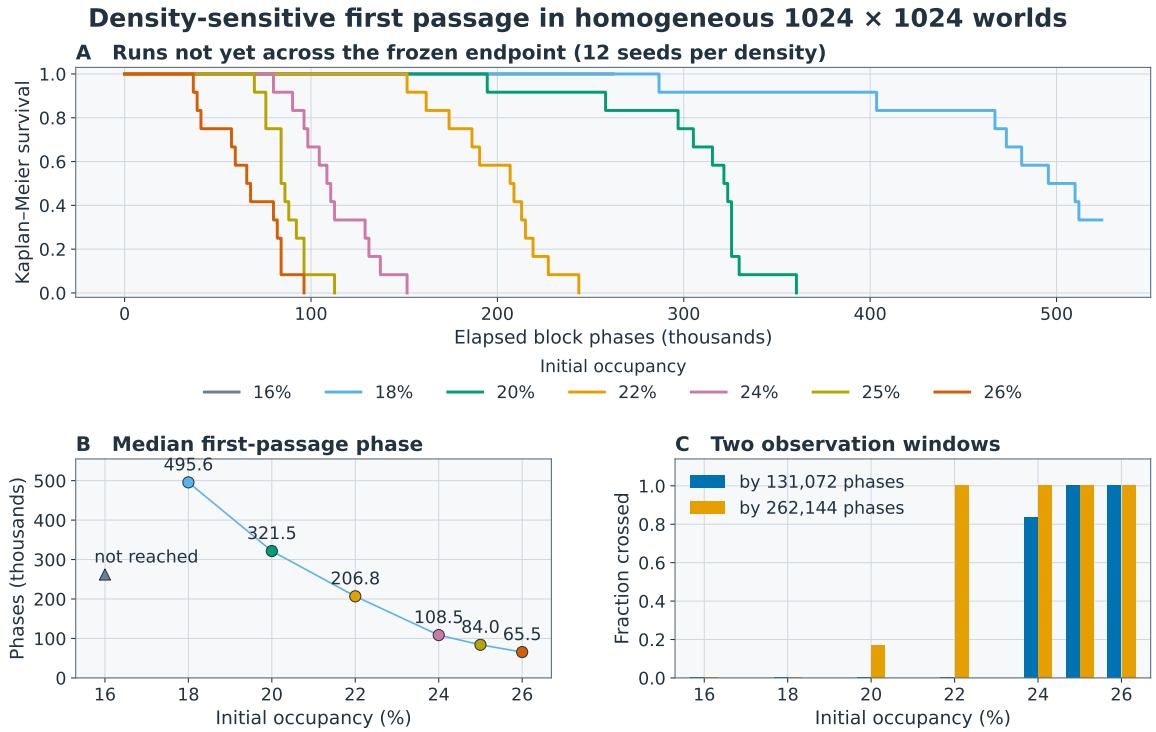


Figure 4: How long the world waits depends strongly on how much material it starts with. A gives the fraction of runs that have not yet crossed the frozen geometrical endpoint against elapsed phases, one curve per initial occupancy as named in the key, 12 seeds per density, with runs that never crossed kept as censored rather than assigned an event time. B plots the Kaplan–Meier median first passage against initial occupancy: 495,616 phases at 0.18 falling to 65,536 at 0.26, and no crossing at all at 0.16 within its 262,144-phase follow-up. C scores that same endpoint at two observation windows, 131,072 and 262,144 phases, and shows that a density looks like a threshold only until its runs are watched for longer. The spectral diagnostics are recorded separately and form no part of the endpoint. These are finite-time waiting times under one operational endpoint: they do not locate a thermodynamic critical point, establish an infinite-time threshold, or imply a scaling law.

nomenon. The direct persistence measurements and the relative-growth endpoint must be read separately.

5.4 Conservation and reaction controls constrain the mechanism

The closest reactive sibling is a variant of the winner altered as little as possible so that every block update conserves M as well as Q while keeping nonzero reactions. It remained dynamically active (median across runs of the mean reaction density 0.01160, versus 0.01234 for the winner) but produced 0/8 strong-domain outcomes in the focused 1024^2 screen (8 replicates) and 0/16 end-of-window strong-domain passes in the pooled 512^2 balanced protocol (16 replicates); the winner produced 8/8 and 10/16 respectively. A pure-rotation control also produced 0/16. Thus inactivity alone does not explain the sibling result, and transport alone is insufficient under this protocol.

Single reaction-pair removals were heterogeneous. Six of twelve tested pair removals produced no end-of-window pass in any of their 16 replicates; the other removals retained 2–7 end-of-window passes. These pooled counts are taken at the end of the observation window and are not first-passage counts: a run can cross the endpoint and later fall back below its final scale-ratio condition, which happened to one run each in two of the ablations. The architecture is therefore distributed across several channels rather than reducible, from these data, to one uniquely necessary pair. Table 2 and Figure 5 report the full pooled comparison.

Table 2: Matched controls and single-pair ablations: end-of-window strong-domain passes under the pooled 512^2 protocol, 16 replicates each.

Condition	Strong-domain passes	Trials	Interpretation at tested protocol
Frozen winner (exact Q)	10	16	reference
Reactive exact-Q+M sibling	0	16	active, no winner regime
Pure rotation	0	16	transport-only negative control
Ablations 01, 11	7 each	16 each	partial retention
Ablation 05	6	16	partial retention
Ablations 08, 09	3 each	16 each	strong attenuation
Ablation 06	2	16	strong attenuation
Ablations 00, 02, 03, 04, 07, 10	0 each	16 each	suppression

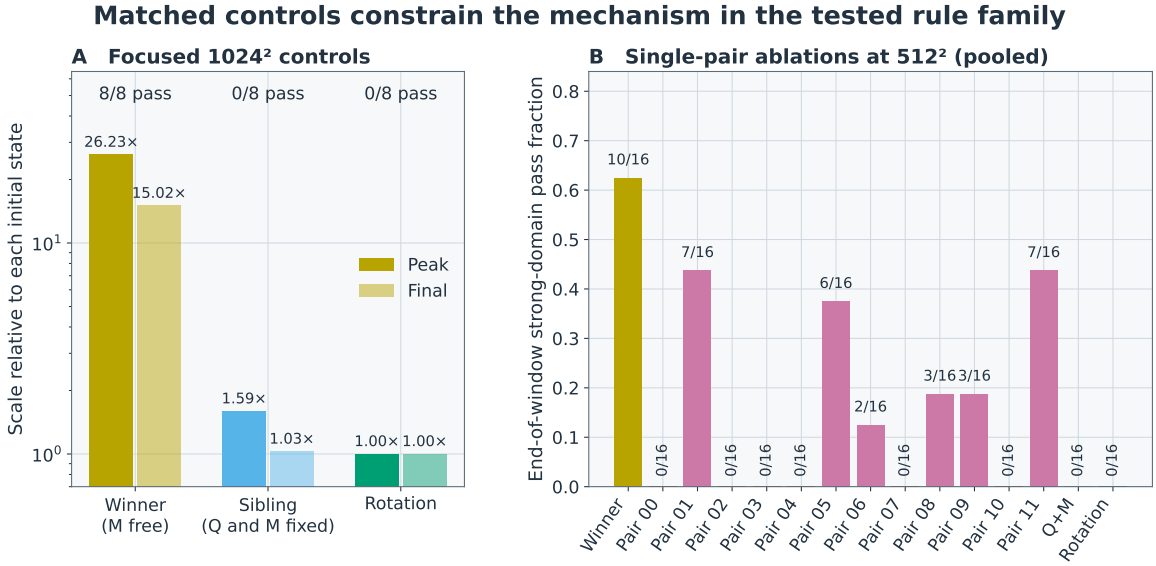


Figure 5: Nearby rules do not do what this one does. A compares the frozen winner with its closest reactive sibling, which conserves occupancy M as well as charge Q, and with pure block rotation, under the focused 1024^2 protocol: the bars give, for each rule, the median across its eight replicates of the run's peak and final finite empty-domain scale relative to its own initial state, and the counts above them are 8/8 strong-domain outcomes for the winner against 0/8 for the sibling and 0/8 for rotation. B is a separate pooled 512^2 protocol with 16 replicates per condition, giving the end-of-window pass fraction for the winner, which passed 10 of 16, for each of twelve single reaction-pair removals, and for the same two controls, which passed none. Six of those removals passed in none of their 16 runs while the rest retained between 2 and 7, so the architecture is spread across several reaction channels rather than resting on one. The two panels use different lattice sizes and replicate counts and should not be read as a single experiment. None of this proves that conserving both Q and M forbids organisation in general, that Q causes the organisation, or that any one reaction pair is universally required.

6. Discussion

The experiments support a compact statement: a fully specified invertible block rule, restricted to the globally charge-balanced sector, can reorganise random material into enormous irregular structures after broad, density-dependent waiting times, while remaining exactly reversible cell for cell. The endpoint is accompanied by substantial long-wave growth but high local compositional entropy. Distributed event morphology, an active exact-Q+M sibling, transport-only controls, and reaction-pair ablations narrow the plausible mechanism without reducing it to a single reaction or established thermodynamic theory.

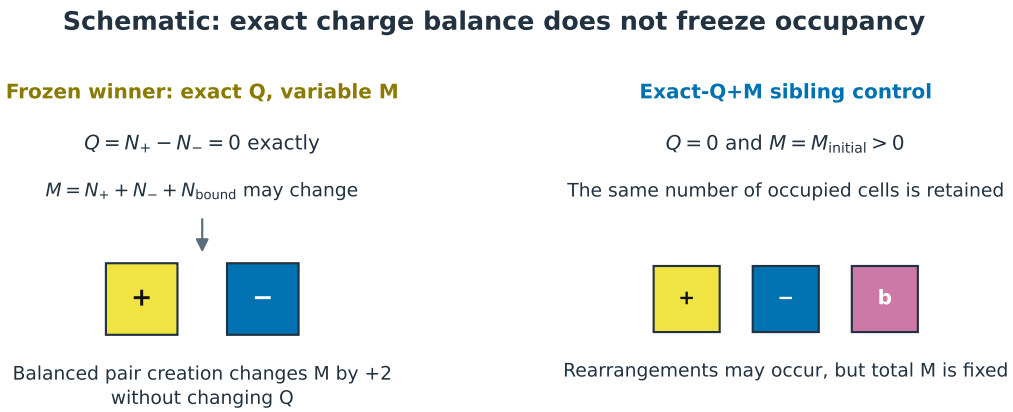


Figure 6: Fixing the net balance does not fix how much material there is. This is a schematic of the two rules' local bookkeeping, not a snapshot of any trajectory. On the left, the frozen winner leaves $Q = N_+ - N_-$ unchanged in every one of its 256 local cases, so a world that begins at zero stays at zero, while $M = N_+ + N_- + N_{\text{bound}}$ is free to move: creating a balanced positive–negative pair raises M by two and leaves Q untouched, and it is that freedom the trajectory measurements then probe. On the right, the exact-Q+M sibling is built so that both quantities are exact; its cells may rearrange, but the number of occupied cells never changes, and under the tested protocol it does not reproduce the winner's strong-domain regime. The comparison is between two specific rules under one finite protocol: it does not establish a general theorem that double conservation forbids organisation.

The closest precedents are substantive. Partitioned reversible automata supply the architecture [3, 24]; Q2R work supplies conservative reversible dynamics with macro-level relaxation [13, 14]; reversible aggregation and reversible lattice gases demonstrate aggregation and apparent macroscopic irreversibility under invertible microscopic rules [16, 25]; one-dimensional two-species reversible automata supply especially close vacancy, species, reaction, and conserved-charge precedents [17, 18]; and reactive lattice gases and reaction automata supply established pattern-forming contexts [10, 19, 26]. The residual contribution is the audited combination of a two-dimensional four-state signed invariant with variable occupancy, delayed distributed coarsening-like organisation, exact large-lattice reversal, and a frozen hierarchy of density, geometry, sibling, and reaction controls. The author-retained search log records query families, databases and publisher checks, and a stopping rule; it is a focused, bounded novelty review rather than proof of exhaustive priority.

The broad waiting-time distribution invites comparison with rare-event and first-passage analysis [22, 27]. Yet the data do not identify a free-energy barrier or stationary rate. Likewise, the separation between exact Q and variable M resembles, only at a conceptual level, theories coupling conserved and nonconserved fields [20, 21]. No equilibrium measure, temperature, or dynamic universality class has been demonstrated.

7. Limitations and withdrawn inference

The v0.3.2 history-erasure headline Withdrawn inference

The later v0.3.2 history-erasure campaign cannot support its original causal headline. Global shuffling removed more than 99% of the source's long-wave power. Yet the fixed endpoint classified many of the shuffled states as "strong domain" immediately or within a few thousand phases, because its connected-empty-cluster terms were confounded by the random high-occupancy geometry: in the primary cohort shuffled 32,768 phases before each source's own first hit (Branch B, 16 replicates), 9 hit at elapsed zero and 14 by 4,096 phases. The intervention audits and raw trajectories remain valid records, but the first-passage contrast is not a valid assay of preserved causal spatial memory. We therefore withdraw claims that the experiment showed a necessary hidden multiscale precursor or history-inhibition effect. Full diagnostics and the immutable historical report are separated in the Supplement and erratum.

More generally, the study uses one selected winner, finite two-dimensional tori, finite horizons, operational morphology thresholds, and a bounded generator prevalence sample. Three readings that the vocabulary might invite are not supported by these finite-time data: that the density dependence is an equilibrium phase transition with a critical density; that the reversible yet directional-looking histories demonstrate a thermodynamic arrow; and that conserving Q is what causes the organisation, rather than a constraint under which it happens to occur.

For each fixed torus, the configuration together with the partition phase evolves by a permutation of a finite state space. Microscopic trajectories are therefore periodic, although their periods were not measured here; the reported coarsening-like behaviour is a finite-time macroscopic episode, not an asymptotic attracting phase.

8. Reproducibility and evidence integrity

The release records the raw-uint8 winner hash, exact configurations, seeds, horizons, endpoint definitions, compact CSV summaries, state hashes, figure-source hashes, code version, Windows/CUDA/PyTorch environment, and an independent parity audit. For that audit the frozen rule and the Margolus update were re-implemented in Eshkol, a Scheme-based language compiled to native code [28]: the generated program re-runs the exhaustive 256-entry audit (output range, bijectivity, exact local charge) and replays a 16×16 exact-inventory torus for 256 steps, agreeing with the Python simulator cell for cell and in exact Q and M. Full-resolution state arrays, checkpoints and complete trajectories are retained by the author and are not part of the public release; analyses that need them require the retained data or a new run. Figures are supplied in the formats listed in the release manifest, which records the size and SHA-256 digest of every included file. The original historical v0.3.2 records are retained unchanged by the author; where public text copies were sanitised, the release's `REDACTIONS.md` records those changes, and the additive erratum and corrected derivative remain distinct from the historical account.

This manuscript is version 1.37, the publication version. Earlier revisions (0.2.1 to 0.2.7 and 1.27 to 1.36) corrected methods and provenance descriptions, disclosure, figures, typesetting and citations without changing the rule, detector, stored results or historical experimental records; no new scientific simulation was run for any of them, and none is run for this version. Figure 1 is a presentation-only rearrangement of the same archived image panels, not a new simulation or a new selection of representative states.

9. Code and data availability

A versioned archival deposit accompanies this paper, and a public repository carries a curated copy of it. The release includes the frozen rule tables and controls; forward and inverse simulation code; analysis and figure-generation code; tests and environment specifications; configurations, seeds and preregistration records; compact trajectory summaries and per-run results, including failures and censored runs; the endpoint-correction record; and provenance records for the representative states shown in the figures; the state arrays themselves are retained by the author. These materials should support both independent scrutiny of the reported statistics and regeneration of the experiments without requiring all full-length trajectories to be stored in Git.

Archived code and data release: Zenodo, DOI 10.5281/zenodo.22270456 (this revision).

Public code repository: github.com/BartlettCJ/zws-reversible-coarsening, a curated copy of the release (code, evidence tables, paper sources and figures). The archived Zenodo release remains the citable, hash-pinned source for code, data and provenance.

The release manifest identifies the exact archived version, included files and any omitted large states or trajectories. Regeneration instructions distinguish what can be reproduced directly from the compact release from what requires a new simulation run. No new scientific simulations were run for the present manuscript revision.

10. Conclusion

In plain terms, this artificial world builds large, irregular structures out of random material without ever upsetting its overall plus–minus balance, and its microscopic evolution can be reversed cell for cell. The structures need not be crystals: the dense regions retain high local compositional disorder.

What matters is not the arithmetic of cancellation. It is that a simple, fully specified rule produces a reproducible pattern-forming regime, with long delays that depend strongly on starting density and outcomes that change under targeted reaction controls. Fixing the net balance does not fix the amount of occupied material, its arrangement, or how those arrangements change. The active double-conservation sibling and the reaction-pair removals constrain this particular mechanism; they do not show that signed balance causes organisation or that changing occupancy is necessary in every reversible system.

The study therefore moves beyond a static example of quantities summing to zero to an experimentally characterised example of organised dynamics under that constraint.

Acknowledgements and AI-assistance statement

Unlike the parent paper, *Zero Without Stasis*, this companion paper is the product of a mostly autonomous AI research programme, and it is described as such deliberately. Chris Bartlett supplied the motivating idea (its signed-refinement conjecture), set the objective of testing it computationally, provided the hardware, and approved the scope and the stopping rules. He did not design the experiments, write the software, run the analyses, or draft the text. Those were done by AI systems: the experimental protocols, package releases and manuscript drafts by OpenAI's ChatGPT; the simulation campaigns and repository work by OpenAI Codex agents running unattended on the author's machine; and independent critical review by Anthropic's Claude models and by OpenAI's models. The author's part was to read the stage summaries,

approve or redirect each next stage, commission the reviews, decide when to stop, and deposit the release; he accepts responsibility for the claims, interpretations and any remaining errors. Supporting project and review records are retained by the author. The independent parity oracle is written in Eshkol, a Scheme-based language by tsotchke released under the MIT licence [28]; its exhaustive rule audit and cell-for-cell replay are recorded in the release.

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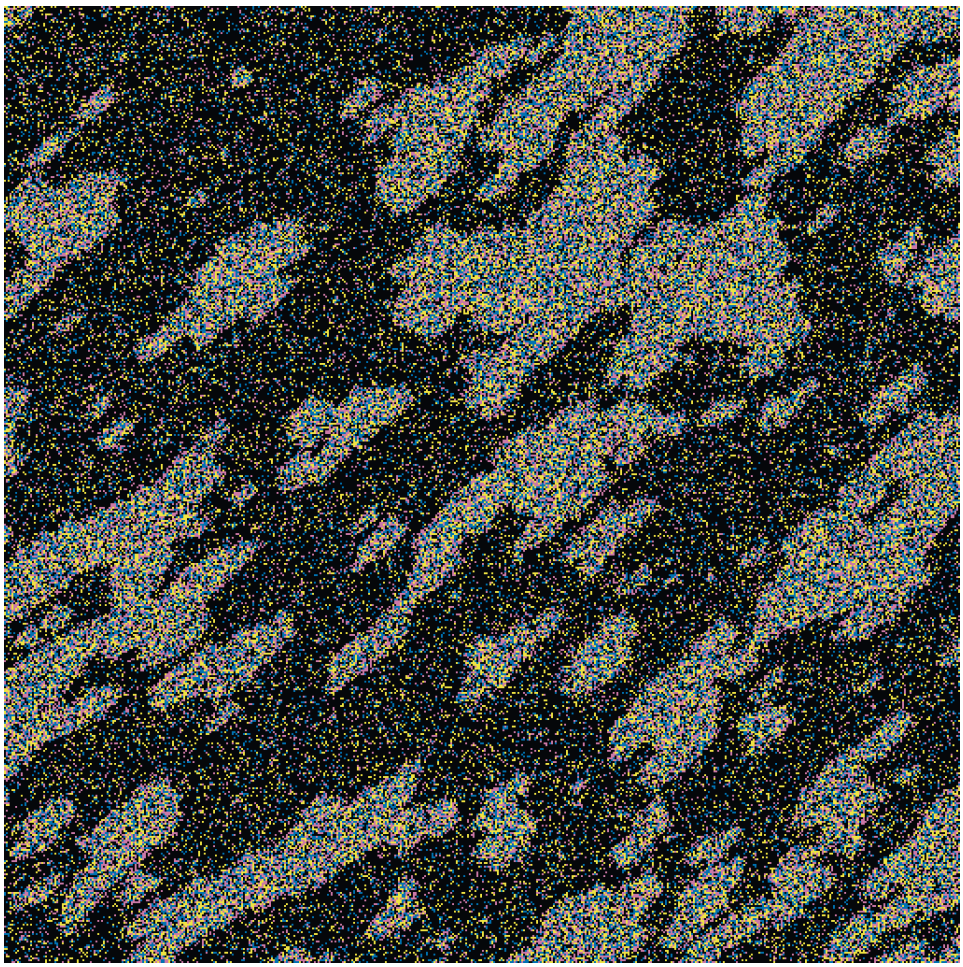
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Zero, without stasis.

This world began as randomly scattered material, and at every step its plus and minus cells have cancelled to exactly zero. Yet it has sorted itself into large islands of busy material and empty channels: order at the large scale, made from a mixture that stays highly varied up close, and every step of that history could be run backwards exactly. A 512×512 window of the 1024×1024 world shown in Figure 3 (initial occupancy 0.20, seed 6203, phase 311,296), rendered cell for cell from the archived state: yellow positive, blue negative, pink bound-neutral, black empty. Global $Q = 0$; $M = 431{,}677$ of 1,048,576 cells.

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